

Employee Turnover Analytics

Portobello Tech - ML Project Report

Dataset Size	14,999 rows - 11,991 after deduplication
Target Column	left (0 = stayed, 1 = left)
Class Balance	83.4% stayed 16.6% left (imbalanced)
Best Model	Gradient Boosting (Accuracy: 98.07%)

1. Executive Summary

Portobello Tech sought to predict and prevent employee turnover using HR evaluation data. After cleaning 14,999 records and handling class imbalance via SMOTE, three ML models were trained using 5-Fold Cross Validation. Gradient Boosting was selected as the best model with 98.07% accuracy and the highest recall for identifying employees likely to leave. Key finding: low satisfaction is the strongest driver of turnover ($r = -0.35$). Using the model's probability scores, 2,399 test employees were categorised into 4 risk zones with targeted retention strategies assigned to each.

2. Data Wrangling

Initial quality check performed on 14,999 employee records across 10 columns.

Check	Result	Action Taken
Missing values	None	No action required
Duplicate rows	3,008	Removed - kept first occurrence
Data types	8 numeric, 2 string	sales & salary encoded via get_dummies
Class balance	83.4% / 16.6%	Handled using SMOTE upsampling

Observations

- 8 numeric columns: float64(2), int64(6)
- 2 categorical columns: salary (ordinal) and sales (nominal)
- left, Work_accident, promotion_last_5years contain only 0 and 1
- No missing values found across all columns
- 3,008 duplicate rows removed - clean dataset: 11,991 rows
- Target column is imbalanced: 83.4% stayed, 16.6% left

3. Exploratory Data Analysis

3.1 Correlation Matrix

- satisfaction_level is highly negatively correlated with left (-0.35)
- time_spend_company is positively correlated with left (0.17)
- number_project is positively correlated with average_monthly_hours, time_spend_company, last_evaluation

3.2 Distribution Analysis (KDE Plots)

Three distinct employee profiles emerged from the satisfaction, evaluation and monthly hours distributions:

Profile	Evaluation	Avg Monthly Hours	Likely Reason for Leaving
The High-Performer "Burnout"	0.8 - 1.0	250 - 300 hrs	Unsustainable workload or better external offer
The Disengaged "At-Risk"	0.4 - 0.5	125 - 150 hrs	Under-utilized, mentally checked out before leaving
The Comfortable "Safe-Zone"	0.6 - 0.8	175 - 225 hrs	Most likely to stay - balanced workload and satisfaction

3.3 Project Count vs Left

- Employees who left handled a slightly higher average number of projects
- Average project count is very consistent across both groups

4. Clustering of Employees Who Left

K-Means (k=3, k-means++, max_iter=500) applied on satisfaction_level and last_evaluation for 1,991 employees who left.

Cluster	Satisfaction	Last Evaluation	Avg Monthly Hrs	Label & Interpretation
0	0.41	0.52	~149 hrs	Mediocre & Checked Out - neither happy nor performing well. Felt stuck with no growth or recognition.
1	0.81	0.91	~216 hrs	Opportunistic Leavers - happy AND high performing but left for better external opportunity.
2	0.11	0.87	~262 hrs	Burned Out High Performers - overworked, underpaid or underappreciated despite delivering results. Most critical to retain.

5. Class Imbalance - SMOTE

	Class 0 (Stayed)	Class 1 (Left)
Before SMOTE	7,999	1,593
After SMOTE	7,999	7,999 ✓ balanced

Observations:

- Categorical columns (sales, salary) encoded using get_dummies with drop_first=True
- Stratified train/test split: 80/20 with random_state=123
- Train: 9,592 rows | Test: 2,399 rows
- SMOTE applied only on training data - test set kept original distribution

6. Model Training - 5-Fold Cross Validation

KFold(n_splits=5, shuffle=True, random_state=123) applied on training data. cross_val_predict used to generate out-of-fold predictions for classification reports.

Model	Accuracy	Precision (1)	Recall (1)	F1 (1)
Logistic Regression	83.38%	~0.56	~0.18	~0.27
Random Forest	98.39%	~0.97	~0.96	~0.97
Gradient Boosting - Best	98.07%	~0.96	~0.97	~0.97

Observations

- Logistic Regression underperforms (83.38%) - linear model cannot capture complex non-linear turnover patterns
- Random Forest achieves 98.39% - strong precision but slightly lower recall for class 1
- Gradient Boosting achieves 98.07% - highest recall for class 1, making it the best model for this problem

7. Best Model & Evaluation Metric

7.1 Confusion Matrix - Gradient Boosting

	Predicted: Stay	Predicted: Left
Actual: Stay	TN = 7,938	FP = 61
Actual: Left	FN = 115	TP = 1,478

7.2 Recall vs Precision - Why Recall Matters

Metric	Error Type	Business Impact	Verdict
Low Precision	False Positives (61)	HR talks to employees who were not leaving	Low Cost - employee feels appreciated
Low Recall	False Negatives (115)	Employee leaves unexpectedly - costs 1.5 - 2x annual salary	High Cost - exact scenario company wants to avoid

Conclusion: Recall is the primary evaluation metric. Higher recall = fewer missed leavers. Gradient Boosting has the highest recall for class 1 and is therefore the best model.

8. Retention Strategies

Gradient Boosting model applied on test set (2,399 employees). Probability scores binned into 4 risk zones.

Zone	Employees	Recommended Actions
Safe Zone - Green (Score < 20%)	~1,986	Appreciate on their work; include in social events and team-building activities
Low Risk - Yellow (20% – 60%)	~280	Quarterly feedback on work-life balance; provide training/workshops; assign as mentors to new hires
Medium Risk - Orange (60% - 90%)	~98	Public recognition or performance-based rewards; clear 6-month promotion roadmap; offer flexible/remote work
High Risk - Red (Score > 90%)	~35	Immediate 1-on-1 meeting to understand frustration; high-value retention bonus (12-month tie-in); reduce project load immediately

9. Summary

Stage	Key Outcome
Data Wrangling	3,008 duplicates removed; no missing values; 11,991 clean records
EDA	Satisfaction is the strongest predictor of turnover ($r = -0.35$)
Clustering	3 cohorts: Burned Out (cluster 2) is the most critical group to retain
SMOTE	Training data balanced from 7999:1593 - 7999:7999
Best Model	Gradient Boosting - 98.07% accuracy, highest recall for leavers
Key Metric	Recall - minimising false negatives is the business priority
Risk Zones	4 zones defined; 133 employees in medium + high risk requiring immediate action